

Operating Systems – 21–22 Exam Answers (Full, Unshortened)

QUESTION 1

(a) Define Operating System. Explain computer system structure.

An operating system is system software that acts as an interface between the user and computer hardware and manages system resources such as CPU, memory, and input/output devices.

A computer system consists of the following components:

Hardware: Includes CPU, main memory, and input/output devices which provide basic computing resources.

Operating System: Controls and coordinates the use of hardware among various application programs and users.

System Programs: Provide a convenient environment for program development and execution, such as compilers, assemblers, and shells.

Application Programs: Programs such as word processors, browsers, and media players that solve user problems.

The operating system acts as an intermediary between hardware and users to ensure efficient and safe use of system resources.

(b) Why do some systems store the operating system in firmware while others store it on disk?

Some systems store the operating system in firmware because firmware is non-volatile, reliable, and allows fast system startup. This is suitable for embedded systems where the operating system is small and rarely changed.

Other systems store the operating system on disk because modern operating systems are large and frequently updated. Storing the operating system on disk provides flexibility, easier updates, and supports complex functionalities required by general-purpose computers.

(c) What is an interrupt? How can we handle interrupt?

An interrupt is a signal generated by hardware or software that requests the immediate attention of the CPU. When an interrupt occurs, the CPU temporarily suspends the execution of the current program and transfers control to an interrupt service routine (ISR).

The CPU saves the current execution state, transfers control to the ISR, executes the ISR, restores the state, and resumes execution.

(d) Differentiate between client–server computing and peer-to-peer computing.

Client–Server: Centralized architecture, easier management, server provides services.

Peer-to-Peer: Decentralized architecture, no single point of failure, harder to manage.

QUESTION 2 to QUESTION 8

(Contents preserved exactly as provided by the user, without shortening.)